

SUMMARY

This video explains the theory of Cartesian products and relations in Discrete Mathematics. The core transition is from set theory to formal relations defined as subsets of Cartesian products.

Key highlights:

Relations are subsets of Cartesian products.

Ordered pairs preserve order.

Total number of relations on a set of size n is $2^{(n^2)}$.

Classification includes reflexive, symmetric, anti-symmetric, asymmetric, and transitive relations.

I. MATHEMATICAL FOUNDATION: CARTESIAN PRODUCT

1.1 Definition

For sets A and B :

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Ordered Pair Property

$(a, b) \neq (b, a)$ unless $a = b$.

Non-Commutativity

$A \times B \neq B \times A$ in general.

Exception

They are equal only if:

$A = B$

OR at least one set is empty.

Empty Set Case

If $A = \emptyset$ or $B = \emptyset$

Then $A \times B = \emptyset$

1.2 Cardinality

If $|A| = m$ and $|B| = n$

$$|A \times B| = m \times n$$

If relation is on set A and $|A| = n$:

$$|A \times A| = n^2$$

II. DEFINING RELATIONS

A relation from A to B is any subset of $A \times B$.

2.1 Universal and Empty Relations

Universal Relation

$$R = A \times B$$

Empty Relation

$$R = \emptyset$$

2.2 Relations on a Single Set

$$R \subseteq A \times A$$

Total number of possible relations:

$$2^{(n^2)}$$

III. CATEGORIZATION AND PROPERTIES OF RELATIONS

3.1 Diagonal (Identity) Relation

$$\Delta A = \{(a, a) \mid a \in A\}$$

There is exactly one diagonal relation.

$$|\Delta A| = n$$

3.2 Reflexive and Irreflexive Relations

Property	Logical Definition	Constraint
Reflexive	$\forall a \in A, (a, a) \in R$	All diagonal elements present
Irreflexive	$\forall a \in A, (a, a) \notin R$	No diagonal elements allowed

Reflexive and Irreflexive are not exact opposites.
A relation may be neither.

Counting

Number of reflexive relations = $2^{(n^2 - n)}$

Number of irreflexive relations = $2^{(n^2 - n)}$

Examples

$\leq$ is reflexive

$<$ is irreflexive

3.3 Symmetric, Anti-Symmetric, and Asymmetric Relations

Symmetric

If $(a, b) \in R$ then $(b, a) \in R$

Diagonal elements do not affect symmetry.

Universal and empty relations are symmetric.

Anti-Symmetric

If $(a, b) \in R$ and $(b, a) \in R$ then $a = b$

For $a \neq b$, both cannot coexist.

Asymmetric

If $(a, b) \in R$ then $(b, a) \notin R$

No diagonal elements allowed.

Every asymmetric relation is anti-symmetric.

Not every anti-symmetric relation is asymmetric.

3.4 Transitive Relations

R is transitive if:

If $(a, b) \in R$ and $(b, c) \in R$

Then $(a, c) \in R$

Violation occurs if:

(a, b) and (b, c) exist but (a, c) is absent.

IV. ANALYTICAL CASE STUDY: MULTISSETS

Problem

Number of multisets of size 4 from a set of size 10.

Approach

Classify by number of distinct elements.

Type	Count
4 distinct elements	$C(10,4) = 210$
3 distinct, one repeated	$C(10,3) \times 3 = 360$
2 distinct elements	$C(10,2) \times 3 = 135$
1 element repeated 4 times	$C(10,1) = 10$

Total

$$210 + 360 + 135 + 10 = 715$$

Alternative Method

Combination with repetition:

$$(n + r - 1)C_r$$

$$\begin{aligned} &(10 + 4 - 1)C_4 \\ &= 13C_4 \\ &= 715 \end{aligned}$$